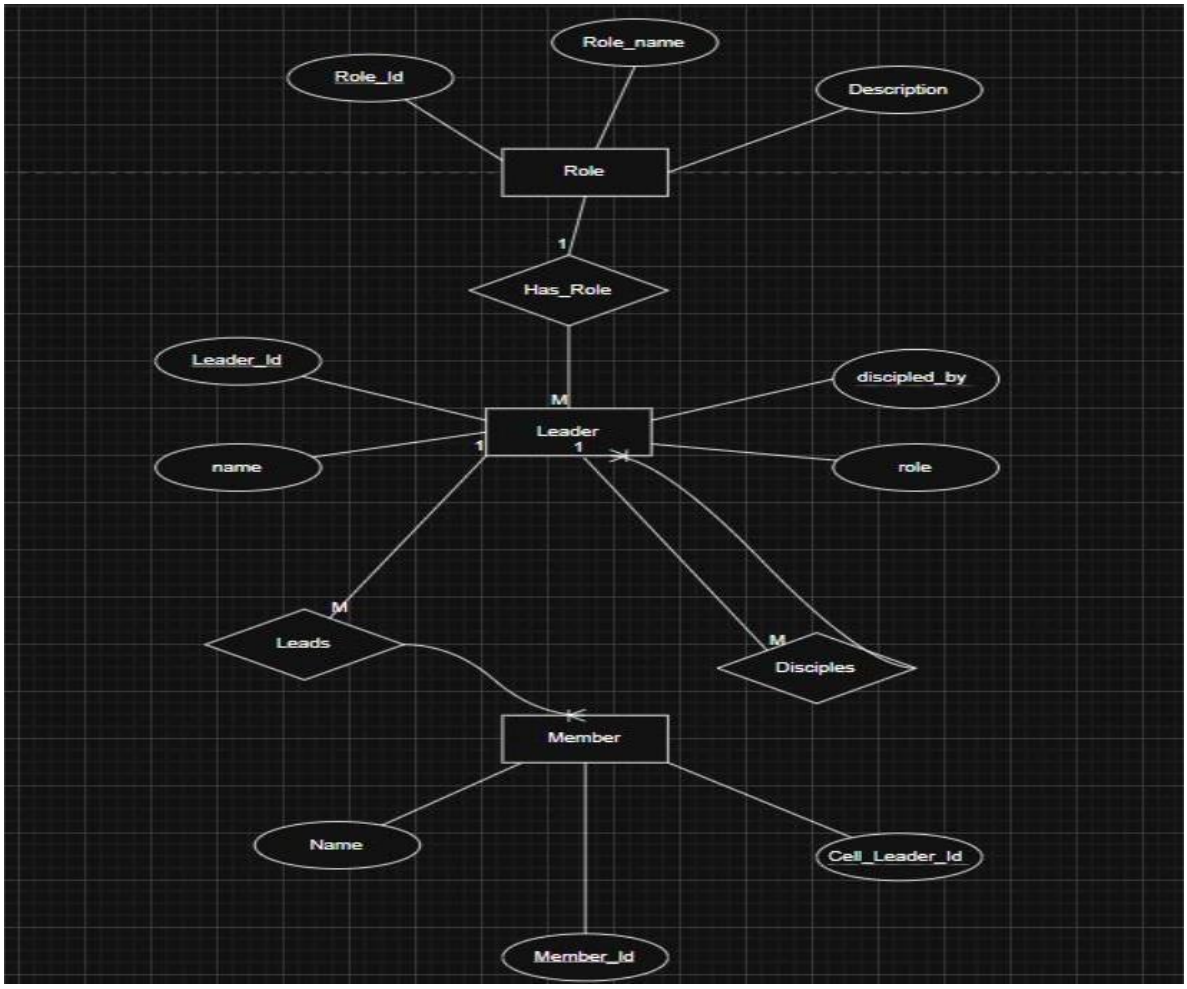
G12 Church Leadership Database Project

Business Rule

The G12 Church leadership system follows a scalable discipleship model where one Senior Leader mentors twelve Primary Leaders, each overseeing groups of 12 consisting of cell leaders and members. Every Leader, except Senior Leader, must be discipled by exactly one person above them. Members belong to a specific cell leader responsible for their development and support. Roles define leadership levels such as Senior Leader, Primary Leader, Cell Leader, clarifying responsibilities and communication pathways. This structure ensures coordination and organised reporting across the church. This structured hierarchy ensures accountability, coordination, and organised reporting across the church.

ERD Description



The entity relationship diagram presented above includes 3 entities being the : Leader, Member, and Role Entities. The leader entity includes a self-referencing relationship through the discipled_by attribute, showing that a leader can lead other leaders if the others leaders are in their cell.

Leaders are then also linked to roles in a one-to-many relationship through the Has_Roles relationship, where each role classifies the leadership level assigned to multiple leaders. The Member entity is then connected to Leader in a one-to-many relationship through the Leads relationship, this is used to indicate that each Cell Leader oversees many members, while each member belongs to exactly one leader. Primary keys and Foreign Keys such as leader_id, member_id, and Cell_Leader_Id maintain relational integrity across the hierarchy.

SQL QUERIES:

All SQL queries were executed on an SQLite relational database, using standard SQL syntax, which is fully compatible with MySQL for the purpose of this coursework

SQL Query 1:

```
SELECT name

FROM Leader

WHERE discipled_by  = 1;
```

Explanation:

This query is used to return all Primary Leaders. The Senior Pastor has an ID of 1, any leader whose discipled_by value that is equal to 1 is one of his Primary Leaders who he mentors.

SQL Query 2:

```
SELECT name

FROM Leader

WHERE role  = 3;
```

Explanation:

This query is used to list all leaders who are only “Cell Leaders”. It enables us to filter through the Leader table based on the role attribute that is within it. It allows us to view everyone who is currently serving as specifically a Cell Leader.

SQL Query 3:

```
SELECT discipled_by, COUNT(*) AS total_disciples
```

```
FROM Leader

WHERE discipled_by  IS NOT NULL

GROUP BY discipled_by;
```

Explanation:

This query would be used to count how many leaders each leader is discipling. It allows you to group the Leader table by the discipled_by field, it then calculates how many leaders are assigned under each leader.

SQL Query 4:

```
SELECT cell_leader_id, COUNT(*) AS member_count

FROM Member

GROUP BY cell_leader_id;
```

Explanation:

This query would be used to count how many church members are led by each Cell Leader. It groups the Member table using the cell_leader_id, it then shows the size of each of the cell groups and how many member each Cell Leader is responsible for.